Mriganka Debnath

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Summary

Backend engineer with five internships building **RESTful APIs, microservices, and serverless systems** in **Node.js, TypeScript, Java (Spring Boot), and Python**. Delivered a **HIPAA-compliant ABHA API** integration for **1,000+ daily active users** and reduced API response times by **35%**. Experienced in relational data modeling on Oracle SQL and Amazon RDS, CI/CD automation, and **AWS** deployment.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, C++, C, SQL
Frameworks: Spring Boot, Node.js, Express, Django, Flask, Next.js, React, Angular
Databases: PostgreSQL, MySQL, Oracle SQL, MongoDB, Amazon RDS
Cloud & DevOps: AWS (Lambda, RDS), Docker, Git, Bitbucket, Jira, CI/CD, Postman
Practices: REST API Design, Microservices, Micro-frontends, Object-Oriented Design (OOD), Test-Driven Development (TDD), API Security, HIPAA Compliance

Professional Experience

Backend Engineering Intern, LabCorp **Feb 2026 – Aug 2026**

- Engineered enterprise services with **Spring Boot** and **Oracle SQL 18**, applying **microservice** patterns so each service deploys independently.
- Coordinated Jira sprint planning and Bitbucket pull-request reviews throughout a **6-month** internship.
- Developed **Angular micro-frontend** components that split delivery into **2 independent release streams** (frontend and backend).

Software Development Engineer Intern, Indian Institute of Technology, Madras **May 2025 – Jul 2025**

- Deployed a **serverless** architecture on **2 AWS services** (**Lambda** and **RDS**) to ingest and store large volumes of environmental data records.
- Launched **peopleswaterdata.org** from scratch within a **3-month** internship, a community platform with live heatmaps for water quality monitoring.
- Optimized geospatial data transfer by generating vector tiles with Tippecanoe after benchmarking **2+ alternative approaches**.

Prism Research Intern, Samsung R&D Institute India **Sep 2024 – Apr 2025**

- Constructed an end-to-end **Python** data pipeline with Hugging Face that converts natural language instructions into 3D scene manipulations.
- Strengthened zero-shot object detection in complex scenes by evaluating and integrating **2 vision models**: **SAM** and **Grounding DINO**.

Backend Developer Intern, Jivan Medical Wellbeings Pvt Ltd **Jun 2024 – Jul 2024**

- Delivered **Node.js, TypeScript, and Next.js** services for **1,000+ daily active users**, reducing API response times by **35%**.
- Integrated **ABHA API V3** in compliance with **HIPAA** for **1,000+ users**, validating endpoints and edge cases with Postman.
- Shipped an AI-powered chatbot in the core application, lifting patient engagement by **40%**.
- Automated API tests and configured **CI pipelines** within a **2-month** internship, shortening release cycles and catching regressions before deployment.

Backend Developer Intern, Chakralaya Analytics Pvt Ltd **Jun 2024 – Jul 2024**

- Implemented **3 ranking algorithms** (AHP, TOPSIS, Best Worst Method) in **Python and Django**, connected to a React frontend for supplier selection.
- Consolidated **3 decision methods** into **1** multi-criteria engine that reduced manual supplier evaluation time and improved selection accuracy.

Projects

Railway Collision Avoidance System GitHub

- Authored **2 core modules**: a **Python/Flask WebSocket** monitoring service that pushes instant safety alerts to dispatchers, and a track allocation engine.
- Prevented conflicts among **2+ trains** in a simulated network using **A* pathfinding** for track allocation.

B2B Online Print Delivery Platform GitHub

- Architected a **Next.js** platform with custom authentication that digitizes ordering of **2 product categories**: large-format engineering blueprints and maps.
- Designed the system architecture and local hosting setup for **3 core workflows**: user authentication, order processing, and high-volume print requests.

Leadership & Achievements

Winner, Best Use of Streamlit, MLH Frost Hacks **2024**

- Won the Streamlit sponsor track among **200 finalists** with a time-series drought prediction tool.

Operations Lead, Bengali Literary Society (Aikyataan), VIT Chennai **2023 – 2025**

- Led a **150+ member** organization and executed **10+ campus events** with **5,000+** combined attendance over two academic years.
- Increased attendance **45%** year over year through targeted multi-channel marketing campaigns.
- Orchestrated cross-functional logistics teams, achieving **100%** on-time delivery across all events.

Education

Vellore Institute of Technology, Chennai, B.Tech in Electronics and Computer Engineering (CGPA: 8.74/10) **2022 – 2026**
Sri Chaitanya Techno School, Visakhapatnam, Senior Secondary, Science (PCM) (Score: 95.6%) **2019 – 2021**